

Article

Current and Potential Land Use/Land Cover (LULC) Scenarios in Dry Lands Using a CA-Markov Simulation Model and the Classification and Regression Tree (CART) Method: A Cloud-Based Google Earth Engine (GEE) Approach

Elsayed A. Abdelsamie ¹, Abdel-rahman A. Mustafa ², Abdelbaset S. El-Sorogy ³, Hanafey F. Maswada ⁴, Sattam A. Almadani ³, Mohamed S. Shokr ^{5,*}, Ahmed I. El-Desoky ⁶ and Jose Emilio Meroño de Larriva ⁷

- ¹ National Authority for Remote Sensing and Space Sciences, Cairo 1564, Egypt; elsayed.abdelsattar@narss.sci.eg
 - ² Soil and Water Department, Faculty of Agriculture, Sohag University, Sohag 82524, Egypt; a_mustafa32@agr.sohag.edu.eg
 - ³ Geology and Geophysics Department, College of Science, King Saud University, P.O. Box 2455, Riyadh 11451, Saudi Arabia; asmohamed@ksu.edu.sa (A.S.E.-S.); salmadani@KSU.EDU.SA (S.A.A.)
 - ⁴ Agricultural Botany Department, Faculty of Agriculture, Tanta University, Tanta 31527, Egypt; hanafey2000@agr.tanta.edu.eg
 - ⁵ Soil and Water Department, Faculty of Agriculture, Tanta University, Tanta 31527, Egypt
 - ⁶ Department of Soils and Water, Faculty of Agriculture, Al-Azhar University, Assiut 71524, Egypt; dr.eldesoky1@gmail.com
 - ⁷ Department of Graphic Engineering and Geomatics, Campus de Rabanales, University of Cordoba, 14071 Cordoba, Spain; ir1melaj@uco.es
- * Correspondence: mohamed_shokr@agr.tanta.edu.eg



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Abstract: Rapid population growth accelerates changes in land use and land cover (LULC), straining natural resource availability. Monitoring LULC changes is essential for managing resources and assessing climate change impacts. This study focused on extracting LULC data from 1993 to 2024 using the classification and regression tree (CART) method on the Google Earth Engine (GEE) platform in Qena Governorate, Egypt. Moreover, the cellular automata (CA) Markov model was used to anticipate the future changes in LULC for the research area in 2040 and 2050. Three multi-spectral satellite images—Landsat thematic mapper (TM), enhanced thematic mapper (ETM+), and operational land imager (OLI)—were analyzed and verified using the GEE code editor. The CART classifier, integrated into GEE, identified four major LULC categories: urban areas, water bodies, cultivated soils, and bare areas. From 1993 to 2008, urban areas expanded by 57 km², while bare and cultivated soils decreased by 12.4 km² and 42.7 km², respectively. Between 2008 and 2024, water bodies increased by 24.4 km², urban areas gained 24.2 km², and cultivated and bare soils declined by 22.2 km² and 26.4 km², respectively. The CA-Markov model's thematic maps highlighted the spatial distribution of forecasted LULC changes for 2040 and 2050. The results indicated that the urban areas, agricultural land, and water bodies will all increase. However, as anticipated, the areas of bare lands shrank during the years under study. These findings provide valuable insights for decision makers, aiding in improved land-use management, strategic planning for land reclamation, and sustainable agricultural production programs.

Keywords: CART; drylands; GEE; LULC; CA-Markov

1. Introduction

Some fertile soils in the Nile Valley and Delta have lost their fertility and production due to deterioration [1]. Food security, which has emerged as the main concern especially in developing country [2], must thus be achieved through the expansion of reclaimed soils and the sustainable development of existing agricultural fields [3]. One of the main

issues with global environmental change and sustainable development is the change in LULC. Rapid changes in LULC have resulted from economic activity and rapid global population growth, which have caused urban agglomeration and subsequent construction land development [4–6]. Poor farming practices and changes in land-use patterns are some of the main causes of soil degradation, nutrient cycling disturbance, and negative impacts on ecosystem services and biodiversity [7].

Because Egypt is an arid country, long-term management and land-use planning of natural resources depend on the identification and projection of LULC changes. The physical makeup and attributes of the land elements on the surface of the earth can be characterized as including LULC. LULC is a multifaceted process with causes and consequences occurring at several social and geographic levels. A basic relationship exists between the dynamic analysis of LULC changes over time and space and large-scale and long-term monitoring methodologies [8,9]. While there is still a comparable demand and supply for land, the need for land is growing. Policies pertaining to LULC must be modified in order to satisfy this need. Researchers, governments, and international organizations have always been interested in mapping LULC. In this context, planning and management of natural resources, research and assessment of environmental quality, and regular monitoring of existing ecosystems are all dependent on the achievement of accurate LULC products at different spatial scales (local, national, and global) [10]. Effective land management techniques and policymaking are made possible by these processes, which aid in understanding the dynamic nature of landscapes [11–13]. Environmental change is acknowledged to be driven by LULC change at all temporal and geographical dimensions. Along with climate change, biodiversity loss, and air, soil, and water pollution, these changes ought to be the top concerns for human populations [14]. Studying how LULC changes over time is therefore vital and important. Thus, it has become increasingly important for academics and policymakers worldwide to monitor and mitigate the detrimental effects of LULC while maintaining the production of necessary resources.

Field surveys and manual visual interpretation served as the primary foundation for early LULC product mapping. Due to its inefficiency and high expense, this approach finds it challenging to meet the requirements for fast and routine LULC mapping and monitoring [15]. The adaptability of remote sensing, GIS, and LULC change models makes them an effective tool for mapping and monitoring LULC changes. In general, the shortcomings of traditional approaches are that they do not account for environmental, social, or demographic circumstances. Satellite imagery has long been one of the main sources of information for the information extraction, analysis, and monitoring of changes in LULC dynamics. Changes on the earth's surface can be identified thanks to the availability of remotely sensed data, which offer a range of temporal, spectral, and spatial resolutions [16]. More and more, precise and trustworthy LULC maps require gathering and analyzing large amounts of satellite data. Due to its great computational complexity, this work requires massive storage capacity, strong processing, and adaptability to utilize several strategies [17]. Furthermore, as classification techniques have advanced, LULC mapping has made extensive use of machine learning algorithms, such as random forest (RF), gradient boosting decision tree (GBDT), and support vector machine (SVM) [18,19]. According to Jin et al. [20], these have created new opportunities for more thorough LULC mapping. They do, however, also provide significant obstacles to the effectiveness of remote sensing (RS) data processing and collecting [19–21]. As evidenced by the Google Earth Engine (GEE), which not only makes full-band and high-intensity image computing feasible but also enables the integration of multisource and multi-scale global remote sensing data, scientists are working to grow and develop satellite remote sensing cloud storage and cloud computing platforms to meet these challenges [22]. One of Google Lab's products, the GEE, employs cloud computing to provide high-performance supercomputer services for raster and vector datasets [17]. In recent years, a big-data cloud-based network computing resource platform called GEE has surfaced and shown itself to be quite competitive. According to Tamimiminia et al. [23], it offers enormous geographic datasets with previously

unheard-of online computational and visual analytical capabilities. Researchers can pre-process or download multi-temporal image data that satisfy filtering requirements using the GEE platform [24]. LULC classification and analysis can then be performed online using a variety of machine learning methods. Because of this, the GEE platform is being used more frequently to produce large-scale, effective LULC products and change-monitoring studies [25]. GEE provides free access to satellite imagery from several sources, including Sentinel, Moderate Resolution Imaging Spectroradiometer (MODIS), and Landsat. The GEE Explorer and GEE Code Editor are the two web-based platforms by which users can utilize its benefits. With GEE Code Editor, users can create more analyses and presentations by using JavaScript or Python code programming, while with GEE Explorer, users can only examine a restricted collection of satellite imagery [26]. GEE is potentially useful for creating continuous long-term LULC maps because it contains a sizable volume of various long-term raw remote sensing observations and dynamic classification algorithms, such as RF models [27,28] and support vector machines (SVMs) [29,30]. The GEE and related Google Earth platforms have been used as pilot sites for a number of studies, including mapping vegetation indices, global urban land clipping, surface water monitoring, land surface temperature, and LULC mapping using Landsat-8 images [31,32].

Many researchers frequently employ the Markov Chain model (MCM), one of the models used to simulate how LULC develops over time. According to Khwarahm et al. [33] the MCM quantifies conversion states and the rate of transfer between various land-use categories. Although the MCM is capable of estimating land use, it lacks spatial explicitness since it is unable to pinpoint the precise location of changes, which is crucial information for understanding the effects of forecasted changes. The primary aim of this study was to develop a CART model on the GEE platform to analyze changes in LULC from 1993 to 2024. The study's main contribution is the development of a map of land cover in Qena Governorate, Egypt, covering the years 1993–2008 and 2008–2024. Additionally, the study analyzed the spatiotemporal changes in LULC from 1993 to 2024 and used spatial modelling CA-Markov to forecast land-use maps for 2040 and 2050.

2. Materials and Methods

2.1. Description of the Research Area

Qena is the third governorate from the southern border of Egypt, following Aswan and Luxor. It is situated between latitudes 25°13' and 26°15' N and longitudes 32°00' and 32°55' E. As seen in Figure 1, it is bordered to the south by the Luxor Governorate, which broke away from Qena in 2010, and to the north by the Sohag Governorate, the New Valley Governorate to the west, and the Red Sea Governorate to the east. The total area of the Qena Governorate is 9565 km² (<http://qena.gov.eg/Default.aspx> accessed on 5 December 2024). The main geographical feature of the Qena Governorate is the 162 km long Nile River, which flows through the valley from south to north and divides the governorate's territory into east and west (<http://qena.gov.eg/Default.aspx> accessed on 5 December 2024). The Qena Governorate has a total population of 3,224,573, with 80.3% living in rural areas and 19.7% in cities. Qena is one of the governorates with the highest population densities. Qena Governorate has a predominantly agricultural economy. The province's main crops include sesame, alfalfa, palm trees, sugar cane, bananas, wheat, maize, vegetables (tomatoes), and a few aromatic and medicinal plants. Sugar cane is the most important crop, followed by wheat in the winter and maize in the summer [34]. The study area is located in an arid region of North Africa, with hot summers and mild winters with little to no rainfall (<https://www.eaaa.gov.eg/Uploads/Reports/Files/20221128214734177.pdf> accessed on 5 December 2024). The average annual rainfall is about 0.7 mm; rain can fall in April, November, and December. The daily mean temperature rises significantly throughout the summer months. The daily average temperature is 25.9 °C, with lows of 17 °C and highs of 34.1 °C. The average annual evaporation rate is 19.5 mm/day, and relative humidity is 22% lower in the summer than in the winter [35,36]. Table 1 shows the four major

land-use types that comprise the research area: cultivated soils, bare soils, urban regions, and aquatic bodies [34–36].

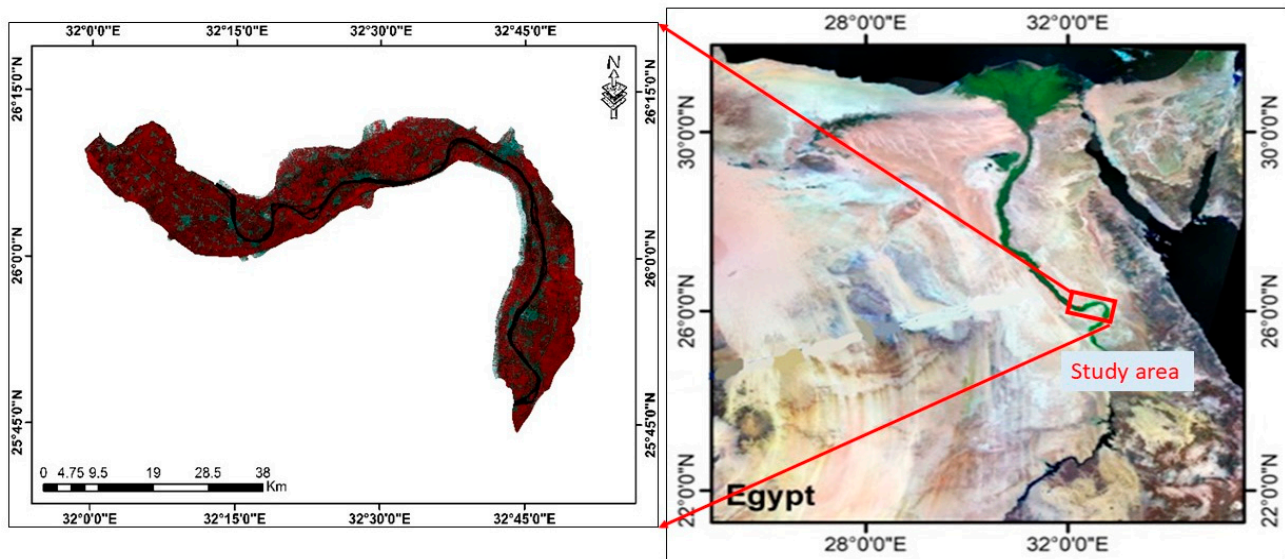


Figure 1. Location of the study area.

Table 1. Description of different land-uses of the study area.

LULC Classes	Description
Urban areas	Urban and rural residential, services, commercial, industrial, and roads.
Water bodies	Nile River, canals, and drainage patterns.
Cultivated soils	Old cultivated soils in the Nile Valley and recently cultivated soil.
Bare soils	Soils of eastern and western parts of Nile Valley.

2.2. The Data Used

For LULC mapping for the years 1993–2024, three multispectral satellite images from Landsat Thematic Mapper TM (15/2/1993–15/12/1993), Enhanced Thematic Mapper ETM+ (15/2/2008–15/12/2008), and Operation Land Imager OLI (15/2/2024–15/5/2024) were used; these images were then compiled and methodically processed on the GEE platform [37]. Additionally, this study employed cloud cover and air pollution to filter the data that were accessible through the downloading data gateway to those with less than 5% cloud cover. The data availability was confirmed using the Google Earth Code Editor platform throughout the acquisition and processing of the satellite images utilizing the GEE platform and the algorithm that was built. Figure 2 presents the flowchart of the approach framework used in the current investigation.

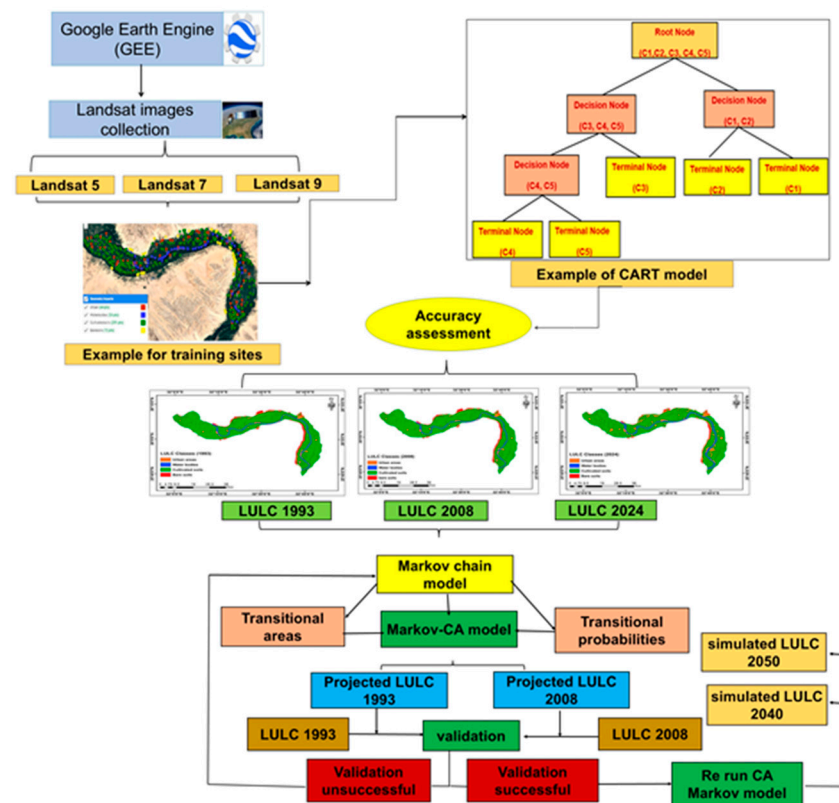


Figure 2. Flowchart of the methodology framework followed in the current study.

2.3. Classification and Regression Tree (CART) Classifier

CART was initially created by Breiman et al. [38] and was based on regression analysis and simple if–else decision-making logic. Through the division of training data into consecutive subgroups (64, 53, 251, and 72 samples for urban areas, water bodies, cultivated lands, and bar soil, respectively), CART constructed a decision tree. Recursive data division based on a predetermined threshold was the result of this phenomenon, which persisted until CART achieved non-dividable leaf nodes on the tree. Afterwards, these leaf nodes on the decision tree were employed as output for forecast, and the decision tree was then utilized in the test phase to evaluate the performance of the algorithm. Because CART is extremely sensitive to changes in the training dataset, even little changes in the training dataset can produce entirely different results. The main drawback of CART is its excessive reliance on training datasets. Even so, CART is a straightforward model that has shown its durability over time. Because of its capacity to manage intricate decision rules and nonlinear interactions, the CART algorithm was selected as the classifier [39]. Preparing training data and utilizing the chosen features are the first steps in training the CART model. A tree-based decision structure is produced by the algorithm’s recursive dataset splitting according to informative attributes [40]. It chooses the most discriminating trait and builds child nodes based on that choice. To optimize the splits, the CART algorithm assesses splitting criteria such as information gain or Gini impurity [41]. After being trained, the CART model can accurately categorize land cover during the winter by applying the decision rules it has learned to new data. This study applied CART classification using the “classifier.smileCart” method that is already integrated into GEE.

2.4. Land-Use Change Analysis by CA-Markov Modeling

The Markov analysis approach is a straightforward statistical tool that relies on a transition probability matrix based on neighborhood effects via a spatially impacted algorithm [42]. The Markov model has gained popularity in recent years as a means of fitting complex spatial patterns using practical rules in the urban development process [43].

Despite its simplicity, the Markov model has a significant restriction regarding the spatial distribution of data related to land uses. Although each land use's transition probability matrix might be precise, it is unknown where the occurrences will occur in space [44]. Therefore, by incorporating a spatial dimension into the model through the application of cellular automata (CA) filter, the CA-Markov chain model was created to improve the accuracy of determining the change of different categories of land use. As a result, CA possesses the potential to change its state based on its neighbors and prior state [45]. Therefore, by first generating a dataset of land uses at a specific time and then projecting the changing probabilities of this data for a future time, a hybrid CA-Markov model can be used to detect at a dynamic degree of estimation the spatial and temporal changes of the various land uses in the Qena Governorate [46]. Nonetheless, the process of detecting LULC changes involves forecasting how the pixels in the satellite image would change over time, switching from one land-use type at time (t_1) to another at time (t_2). For that objective, the Equations (1)–(3) created by Ma et al. [45] were employed.

$$P = (p_{ij}) = \begin{bmatrix} P_{11} & P_{12} & P_{1m} \\ P_{21} & P_{22} & P_{2m} \\ P_{m1} & P_{m2} & P_{mm} \end{bmatrix} \quad (1)$$

$$\sum_{j=1}^m P_{ij} = 1, i, j = 1, 2, \dots, m \quad (2)$$

$$S_{t+1} = P_{ij} \times S_t \quad (3)$$

Here, S_t represents the current land-use status at time t , S_{t+1} refers to the land-use state at the next time point ($t + 1$), and P_{ij} is the transitional probability of shifting from one specific land-use type m to another. The value of P_{ij} ranges between 0 and 1. CA-Markov chain analysis was conducted using IDRISI 7.02 software to determine the transition and probability matrices for specific land-use areas during the periods 1993–2008 and 2008–2024.

2.5. Land Use/Land Cover Change Forecasting

A cellular automaton (CA) with two variables was employed in the simulation procedure for the future land-use forecast. These variables are the local variable, which is typically given interactions, and the discrete variable in time and space [47]. Cell, cell space, rule, neighbor, and time make up this model. Additionally, this model can identify the neighbors; the closer a cell is to its neighbors, the higher the weight factor. As a result, the states of nearby grid cells may be forecasted using the weight factor and transition probability [48]. Equation (4) was utilized to ascertain the change in land use.

$$S_i = \frac{LU_{(i,t_2)} - LUA_i}{LU_{(i,t_1)}} \times \frac{1}{t_2 - t_1} \times 100\% \quad (4)$$

where S_i is land-use change, $LU_{(i,t_1)}$ is land-use change at an earlier time, $LU_{(i,t_2)}$ is land-use changed area at a later time, and LUA_i is area with no change.

The CA-Markov model in the IDRISI 7.02 software was used to estimate the outcomes of LULC alterations in 2040 and 2050. As stated by Tali et al. [49], the LULC images of the Qena Governorate during the two time periods of 1993 as earlier land use (t_1) and 2008 as later land use (t_2) were used in the current investigation.

Using suitable images, the standard contiguity filter (5×5 pixels) was applied to determine the neighborhoods of each land-use class's cells (Equation (6)). A 5×5 matrix space was used to encompass each cell's center in order to produce a good influence. This spatial filter was applied in close proximity to the pre-existing category and excluded arbitrary land-use changes [50]. Additionally, pixels closest to the current land-use category are more suitable than those farther away [51]. In the current work, 2040 and 2050 LULC maps were forecasted using spatially explicit, contiguous weights created from neighboring

pixels. The contiguity filter displayed below is the one utilized for the LULC change analysis (Equation (5)) [3].

$$\text{Contiguity filter } 5 \times 5 = \begin{bmatrix} 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 \end{bmatrix} \quad (5)$$

There are four primary stages to the Markov model's approach for LULC analysis, LULC changes, and forecast. These are as follows:

1. The Markov chains model was used to compute the transition probabilities matrices for the years 1993, 2008, and 2024;
2. From 1993 to 2024, a set of conditional probability data for various land uses was obtained by applying these transition matrices;
3. Using the CA-Markov spatial operator in IDRISI 7.02 software, which is based on Markov chain analysis, the transition probabilities matrices of 1993, 2008, and 2024 as well as conditional probability data were combined to simulate the land-use maps of 2040, 2024, and 2024 as a base map.

A new land-use map was created by overlaying all of the data from the previous steps.

2.6. The Validation Stage of CA-Markov Model Results

The CA- Markov model was validated using the validation module in the IDRISI 7.02 software. The projected outcomes were then contrasted with the observed dataset of LULC changes between 1993 and 2008. This was evaluated by utilizing the kappa index of agreement to estimate the level of agreement between the reference map from 1993 and the simulated map from 2008. The following Equations (6)–(9) represent the kappa indices: kappa for location (k location), kappa for no information (k no), kappa for quantity (k quantity), and kappa for standard (k standard) [52].

$$K_{no} = (M(m)N(n) / (P(p) - N(n))) \quad (6)$$

$$K_{location} = (M(m)N(m) / (P(m) - N(m))) \quad (7)$$

$$K_{quantity} = (M(m)H(m) / (k(m) - H(m))) \quad (8)$$

$$K_{standard} = (M(m)N(n) / (P(p) - N(m))) \quad (9)$$

The degree of conformity between the reference map and the unaltered comparison map is denoted by $M(m)$. The agreement between a map with a distribution of m among the different categories in each cell and the reference map is denoted by $N(m)$. The agreements arising from chance are denoted by $N(n)$. $H(m)$ is the agreement between a modified comparison map and the reference map. The reference map's agreement with a modified comparison map is denoted by $P(m)$; the reference map's agreement with a map that offers perfect information on both quantity and location is denoted by $P(p)$. The agreement between a modified comparison map and the reference map is expressed as $K(m)$ [53].

2.7. Accuracy Assessment

Any classifier's accuracy is influenced by the quantity of input data, the research area, and the satellite sensor. Different LULC accuracies for various classifiers in various study areas have been identified in a number of studies [54–56]. Accuracy assessment can be used to examine the efficiency of any classifier following LULC classification. The kappa coefficient and total accuracy are the most often used metrics for accuracy evaluation. The kappa coefficient evaluates the ability of a classifier in perspective of

the anticipated values. The range of values for the kappa coefficient is 0 to 1, where 0 represents no ability, and 1 represents the classifier's perfect ability. On the other hand, overall accuracy can be defined as the ratio of correctly categorized pixels to all pixels classified. An embedded confusion matrix technique is present in GEE. The GEE-embedded confusion matrix approach was used in this study to validate and assess the classification accuracy. The following Equations (10)–(13) can be used to calculate the kappa coefficient and assess overall accuracy [57].

$$\text{Overall Accuracy} = \left(\frac{C_p}{T_p} \right) \times 100 \quad (10)$$

where C_p indicates correctly classified pixels, and T_p indicates the total pixels.

$$\text{Kappa coefficient} = \frac{N \sum_{i=1}^{cr} x_{jk} - \sum_{i=1}^{cr} (x_j + -x + j)}{N^2 - \sum_{i=1}^r (x_j + \times x + j)} \quad (11)$$

Here, N represents the total number of observations; cr denotes the total count of rows and columns; x_{jk} is the number of observations found in row j and column k ; $x_j +$ is the marginal total for row j ; and $x + j$ is the marginal total for column j . Additionally, the user's accuracy for each class is calculated by dividing the total number of pixels classified in that class by the number of correctly classified pixels in the class. The user's accuracy formula is given as follows:

$$\text{User's Accuracy} = \frac{\text{properly classified pixels of class}}{\text{total classified pixels of class (the row total)}} \times 100 \quad (12)$$

The potential for a pixel to exist in its corresponding projected class is the essence of user accuracy. Accordingly, the ratio of accurately identified pixels to all reference pixels in a class is used to determine the producer's accuracy. The formula for determining producer's accuracy is as follows:

$$\text{Producer's Accuracy} = \frac{\text{properly classified pixels of class}}{\text{total classified pixels of class (the column total)}} \quad (13)$$

2.8. Generation of Thematic Maps

The geospatial analysis wizard provided in the ArcGIS Pro 3.2 software was used to create theme maps of several LULC classes for the studied periods (1993, 2008, and 2024) and forecasted periods (2040 and 2050).

3. Results and Discussion

3.1. Accuracy Assessment

The accuracy assessment of analyzing the LULC change of different land-use classes in Qena Governorate during both time periods (1993–2008) and (2008–2024) indicated that forecasted and observed data are perfectly matched, and a high accuracy of estimation was achieved. The kappa values for these variations range between 0 and 1 (0% and 100%); the closer the value to 100%, the better the accuracy of the agreement [53]. The probability of changing land use from a category for another was also calculated for each land-use class in the studied area using Markov chain modeling of different Landsat satellite images for different years.

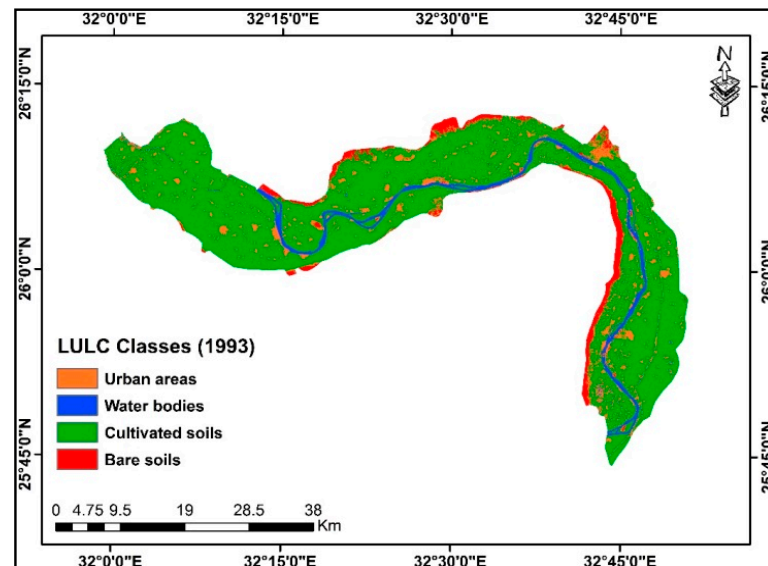
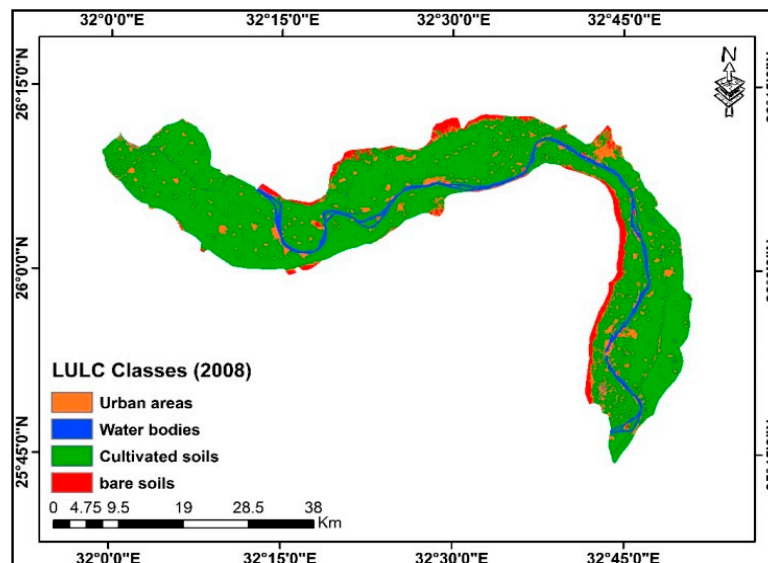
The acquired results, as indicated in Table 2, demonstrate that for the various study periods of 1993, 2008, and 2024, the classification accuracy satisfied the requirement that the sensor data had at least 80% accuracy [54]. The LULC classification's overall accuracy varied between 97.29% and 99.33% (Table 2). These percentages indicate high precision, according to Anderson et al. [55,56]; therefore, the LULC maps can be used reliably for forecasting and studying LULC change dynamics.

Table 2. Evaluation of each classified image’s accuracy from 1993, 2008, and 2024.

Land Uses/Classes	Landsat TM 1993		Landsat ETM+ 2008		Landsat OLI 2024	
	PC (%)	UC (%)	PC (%)	UC (%)	PC (%)	UC (%)
Urban areas	99.94	96.45	99.44	96.27	96.54	97.74
Water bodies	96.43	100	96.61	100	98.52	100
Cultivated lands	100	99.95	99.81	99.93	99.89	99.38
Bare soils	100	99.96	99.18	99.74	90.92	85.06
Overall accuracy (%)	99.33		99.23		97.29	
Kappa coefficient	0.99		0.98		0.96	

Note: PC = producer’s accuracy; UC = user’s accuracy.

Figures 3–5 display the classification maps that were created using the Qena Governorate’s historical images. Different LULC classes were categorized using different and distinguishing colors based on the legends on these maps.

**Figure 3.** The classified image of the study area in 1993.**Figure 4.** The classified image of the study area in 2008.

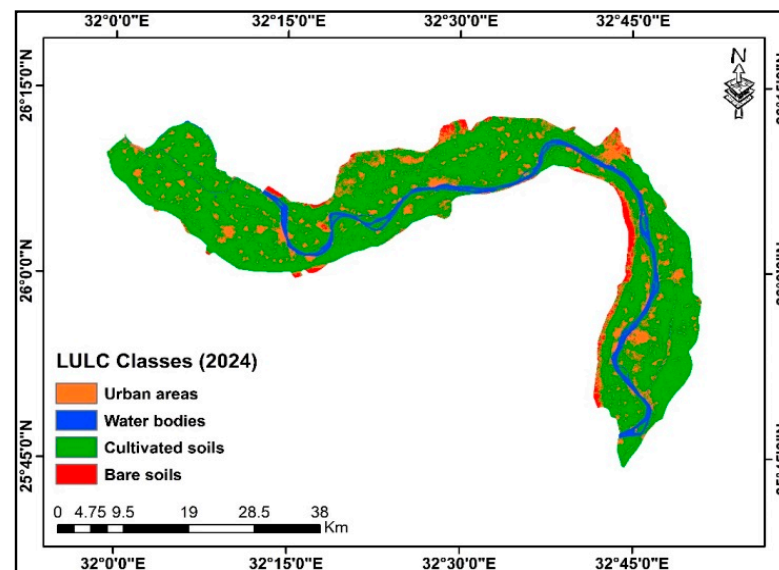


Figure 5. The classified image of the study area in 2024.

3.2. Change Analysis of LULC Maps

Many studies have utilized LULC changes in the past periods and forecasted the possible changes in future situations using remote sensing (RS) and geographic information systems (GIS) together with the CA-Markov model [3,58–61]. The LULC areas' annual change in different classes of the study area was calculated in square kilometers and percentage in the studied time periods. Urban areas spread across both cultivated and bare soils as population grew. Between 1993 and 2008, there was a net growth of 57.00 km² in the urban areas (Figure 6). On the other hand, the net loss for the bare and cultivated soils was 12.4 and 42.7 km², respectively. Actually, during this period (1993–2008), many bare soils were exposed to the reclamation process [34]. As a result, there were more cultivated areas and less bare soil. In terms of water bodies, certain islands entirely vanished within the research region, while other islands sprouted up elsewhere. Depositional processes also caused new islands and sandbars to form as well as new floodplains to arise on the convex sides of the river flow. Therefore, from 1993 to 2008, the Nile River lost 1.9 km² in total.

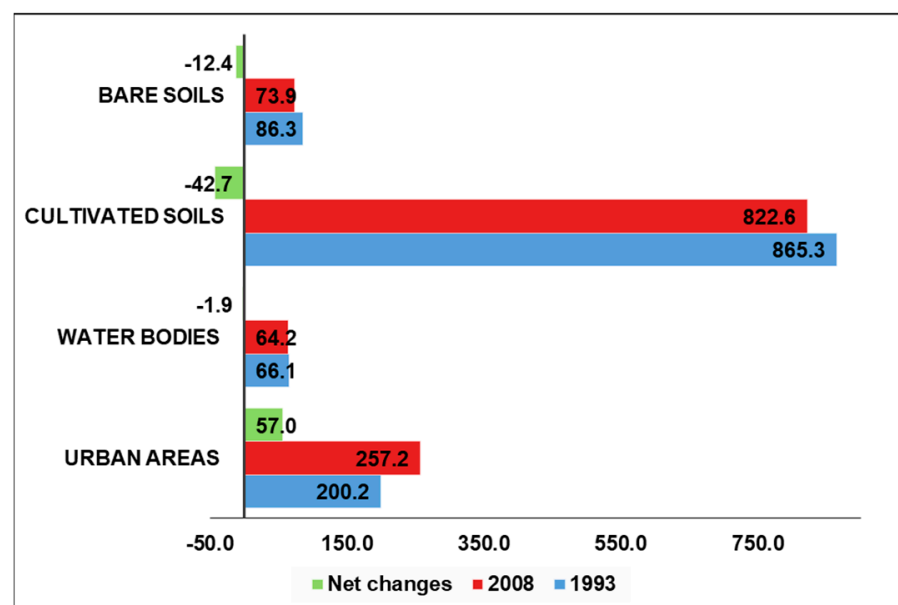


Figure 6. The changes in km² of each LULC category for 1993–2008.

In the research region, the area covered by water bodies increased by almost 24.4 km² over the second period (2008–2024). This could be as a result of numerous canals being built and renovated. Significant advancements have been made in the first phase of the Decent Life program, which was started in 2019 with the goal of promoting rural development, according to the Ministry of Irrigation and Water Resources. As per the ministry's announcement on 7 May 2024, a substantial 3230 km of canals have undergone rehabilitation. It is important to note that the efforts to line the canal are being undertaken in 18 governorates, with specific attention paid to areas like Menya, Luxor, Menoufeya, Asyut, Qena, Beni Suef, Sohag, and Beheira. By ensuring fair access to upgraded water management infrastructure, this planned distribution promotes rural prosperity and crop growth. Reclamation efforts have accelerated during this time, but unfortunately, Qena Governorate is one of the most heavily populated governorates [62]. According to CAPMAS [62], the population of the Qena Governorate increased to approximately 3,224,573 in 2023, with 80.3% of the population residing in rural regions and 19.7% in urban areas. Consequently, the net loss for cultivated and bare soils was 22.2 and 26.4 km², respectively. One of the primary issues endangering the study area's scarce, extremely fertile terrain is urban sprawl. The most productive land in the research region has been significantly lost to urban developments (Figure 7). Built-up areas are rapidly expanding over agricultural soils. The area added to urban settlements was about 24.2 km². The variations in LULC classes in the study area between 2008 and 2024 are shown in (Figure 7).

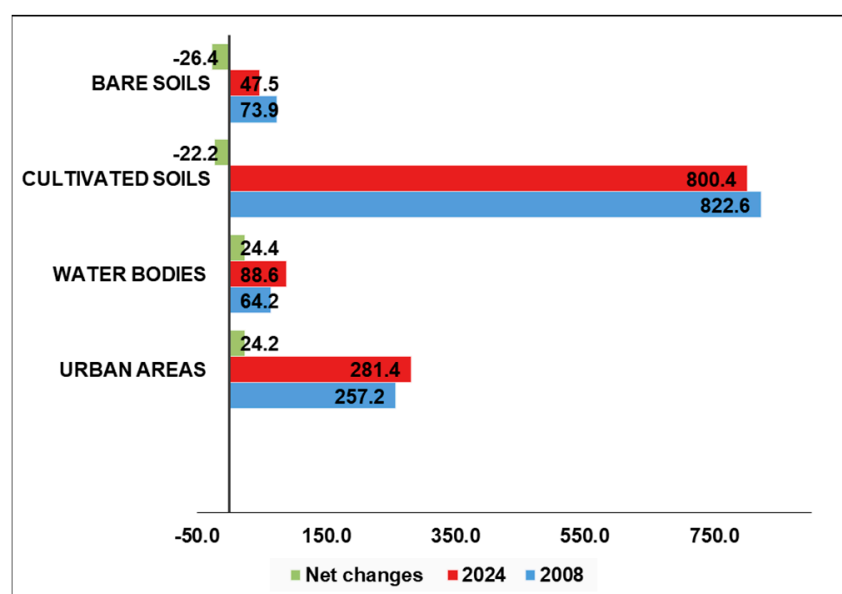


Figure 7. The changes in km² of each LULC category for 2008–2024.

3.3. Markov Chain Model Analysis

The acquired data, which are displayed in Tables 3 and 4, illustrate how the transition probabilities changed over time from 1993 to 2024. Gains are obtained by deducting the persistence (diagonal entries) from each group's total column, while losses are obtained by deducting the persistence from each group's total row [63]. For example, from 1993 to 2008, there was a 52.28% chance that urban regions would remain urban areas; in contrast, there was a 22.44% probability that cultivated soils would eventually become urban areas, and so on for other LULC categories. In 2008, there was a 42.54% and a 76.42% probability, respectively, that the bare soils and water bodies would stay the same.

Table 3. Transition probability matrix based on Qena’s 1993–2008 land-use maps.

Land Uses/Classes	Probability of Changing from 1993 to 2008				Subtotals	
	Urban Areas	Water Bodies	Cultivated Soils	Bare Soils	Total	Loss
Urban areas	0.5228	0.0550	0.413	0.0092	1	0.4772
Water bodies	0.0469	0.7642	0.1889	0.0000	1	0.2358
Cultivated soils	0.2244	0.0396	0.7352	0.0008	1	0.2648
Bare soils	0.4050	0.0014	0.1682	0.4254	1	0.5746
Total	1.1991	0.8602	1.5053	0.4354	4	
Gain	0.6763	0.0960	0.7701	0.0100		

Table 4. Transition probability matrix based on Qena’s land-use maps from 2008 to 2024.

Land Uses/Classes	Probability of Changing from 2008 to 2024				Subtotals	
	Urban Areas	Water Bodies	Cultivated Soils	Bare Soils	Total	Loss
Urban areas	0.839	0.0000	0.161	0.0000	1	0.161
Water bodies	0.1578	0.838	0.0042	0.0000	1	0.162
Cultivated soils	0.1448	0.0097	0.8454	0.0001	1	0.1546
Bare soils	0.1716	0.0000	0.0000	0.8284	1	0.1716
Total	1.3132	0.8477	1.0106	0.8285	4	
Gain	0.4742	0.0097	0.1652	0.0001		

3.4. Land-Use Change Dynamics

Table 5 provides more insight into the LULC class changes of the Qena Governorate area by calculating the annual change for the two analyzed time periods (1993–2008 and 2008–2024). It is clear from the data that were acquired that, for the two time periods, the cultivated lands declined yearly by 2.85 km² and 1.48 km², respectively. The second period in which the reclamation processes of bare soils were observed had the lowest annual reduction. As a result, during the two time periods, the amount of exposed soils decreased by 0.83 km² and 1.76 km², respectively, annually. Over the two investigated periods, the annual growth in urban areas was 3.80 km² and 1.61 km², respectively. In terms of the water bodies, they shrank by 0.13 km² annually between 1993 and 2008 but slowly increased by 1.63 km² between 2008 and 2024 as a consequence of the study area’s canals and waterways being rehabilitated and lined.

Table 5. LULC classes in the Qena Governorate area change annually.

Land Uses/Classes	Average Annual Changes km ²	
	1993 to 2008	2008 to 2024
Urban areas	3.80	1.61
Water bodies	−0.13	1.63
Cultivated lands	−2.85	−1.48
Bare soils	−0.83	−1.76

The results of the LULC change in the various LULC classes in the Qena Governorate across the time periods under study are shown in (Table 6). The temporal distribution of the LULC change in Qena Governorate classes across time, expressed in square kilometers, is shown in (Figure 8). Between 1993 and 2024, urban areas—which include houses, roadways, industrial regions, and other human activities—grew quickly. With population expansion and the broader evolution of the construction of roads for industry and new urban settlements, urban areas have grown significantly, from roughly 200.20 km² (16.44%) in 1993 to roughly 257.20 km² (21.12%) in 2008 and 281.40 km² (23.11%) in 2024. In addition, the urban areas increased clearly [34] (Table 6). People migrated to the low-lying desert

fringe (planned reclaimed lands) areas that were overrun for the limestone plateau from the restricted Nile Valley. The area covered by water bodies in 1993 was roughly 66.10 km² with 5.43%, gradually declining to 64.20 km² (i.e., 5.27% of the entire study area) in 2008 and growing to around 88.60 km² in 2024 as a result of the canals and waterways in the study area being restored and lined. Over the past thirty-one years, the research area's cultivated land has shrunk. Between 1993 and 2024, it underwent a minor alteration, going from 865.30 to 800.40 km², respectively. In 1993, 86.30 km² (7.09%) of the Wadi El Nile area was covered by bare soils, which includes the highlands that encircle the lowlands [34]. This class dropped to 73.87 km² (6.07%) in 2008 and 47.46 (3.90%) in 2024 because of recently proposed land reclamation [34] (Table 6).

Table 6. Land use/land cover distribution across time, expressed in km² by year.

Land Uses/Classes	1993 Areas		2008 Areas		Changed Areas 1993–2008	2024 Areas		Changed Areas 2008–2024
	km ²	%	km ²	%	km ²	km ²	%	km ²
Urban areas	200.20	16.44	257.20	21.12	57.0	281.40	23.11	24.2
Water bodies	66.10	5.43	64.20	5.27	−1.9	88.60	7.27	24.4
Cultivated soils	865.30	71.05	822.60	67.54	−42.7	800.40	65.72	−22.2
Bare soils	86.30	7.09	73.87	6.07	−12.4	47.46	3.90	−26.4

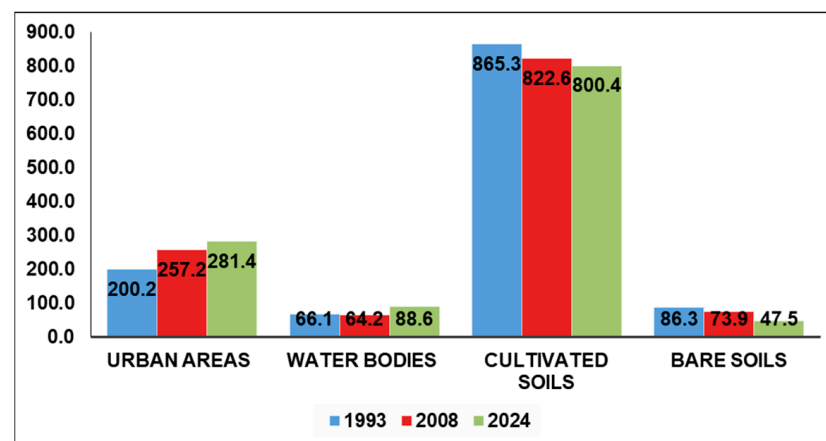


Figure 8. LULC change distribution over time in km² between 1993, 2008, and 2024.

3.5. LULC Changes Prediction of 2040 and 2050

The CA-Markov model's forecast for LULC change across all land-use classes in the Qena Governorate were verified using data from 2008 and 2024. Figures 9 and 10 present the thematic maps of the spatial distribution of the anticipated LULC changes in 2040 and 2050. Table 7 displays the acquired statistics on the anticipated LULC changes in the various classes of the Qena Governorate area throughout the two forecasted time periods (2040 and 2050). In 2040, urban areas will grow by 44.90 km², and by 2050, they will grow to 331.60 km² (27.23%). This anticipated rise in urban regions may be connected to the current population-growth-driven expansion of built-up areas [64–67] (Table 7). Moreover, in 2050, the proportion of water bodies in the study region will rise steadily to 27.90% of its total area. Cultivated lands will drop by 47 km² in 2040 and then reach 731.30 km² in 2050 (Table 7). Urban development will ultimately result in fewer bare soils [3].

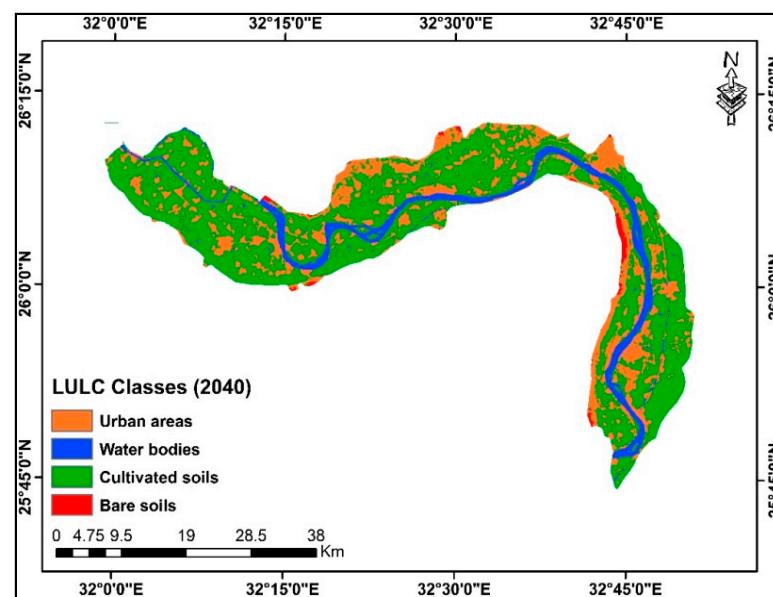


Figure 9. Spatial distribution map of forecasted LULC changes during 2040.

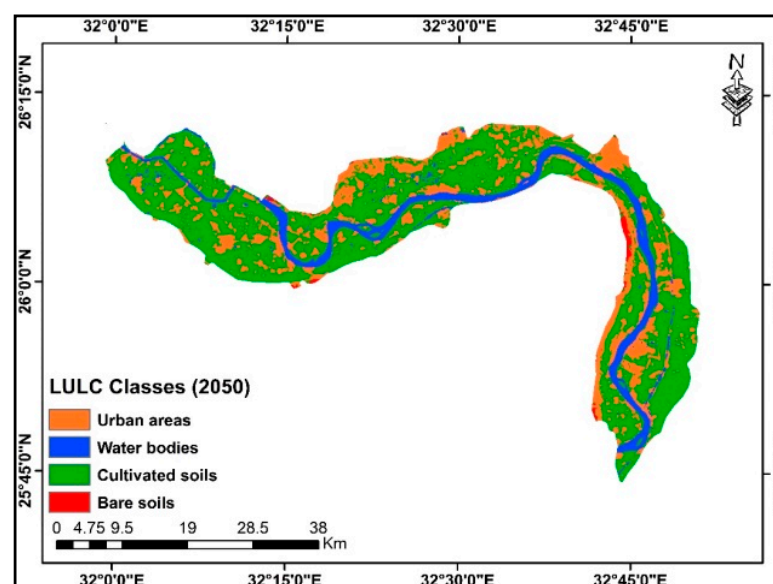


Figure 10. Spatial distribution map of forecasted LULC changes during 2050.

Table 7. Changes in km² that are anticipated for LULC classes in 2040 and 2050.

Land Uses/Classes	Projected	Area	Changed Areas (km ²)	Projected	Area	Changed Areas (km ²)
	2040	%	2024 to 2040	2050	%	2040 to 2050
Urban areas	326.3	26.79	44.9	331.60	27.23	5.32
Water bodies	107.4	8.82	18.8	135.30	11.11	27.90
Cultivated lands	753.4	61.86	−47.0	731.30	60.05	−22.10
Bare soils	30.8	2.53	−16.7	19.70	1.62	−11.08
Total	1217.9	100.00		1217.90	100.00	

3.6. Model Validation

The IDRISI software's validate tool was used to generate the kappa statistics of the k standard, k no, k location, and k location layer to further confirm the tests. Kappa can be used to describe varying levels of consistency. It can be classified into five groups: slight (0.0–0.20), fair (0.21–0.40), moderate (0.41–0.60), substantial (0.61–0.80), and almost perfect (0.81–1). Based on the comparison between the categorized LULC map for 2008 and 2024 and the forecasted LULC for 2008 and 2024, the k coefficient for the quantity and location was determined for the purpose of validating the model. The k standard is 0.9321, k no is 0.9640, k location is 0.9956, and k location layer is 0.9956, according to the statistics of k indicators for 2008. Comparably, the 2024 k indicator statistics indicate that the k standard is 0.9246, the k no is 0.9599, the k location is 0.9504, and the k location layer is equal to 0.9504 (Table 8). As the outcomes, the quantification and localization mistakes were greatly reduced. Both precise location and quantity specification are perfectly attainable with the simulated model; in addition, the projected and real maps had small quantification and location errors [47,68,69]. Based on the good agreement between the anticipated and real LULC map, Table 8 and Figure 11 verify that the accuracy evaluation of the both classified data of 2008 and 2024 is suitable and acceptable for any application. This shows that the number and location of the LULC changes can be exactly specified by using the CA-Markov model [5,68,70,71]. These results completely concur with those of [3,60,72].

Table 8. K indicators for 2008 and 2024.

K Indicators	2008	2024
k standard	0.9321	0.9246
k no	0.9640	0.9599
k location	0.9956	0.9504
k location layer	0.9956	0.9504

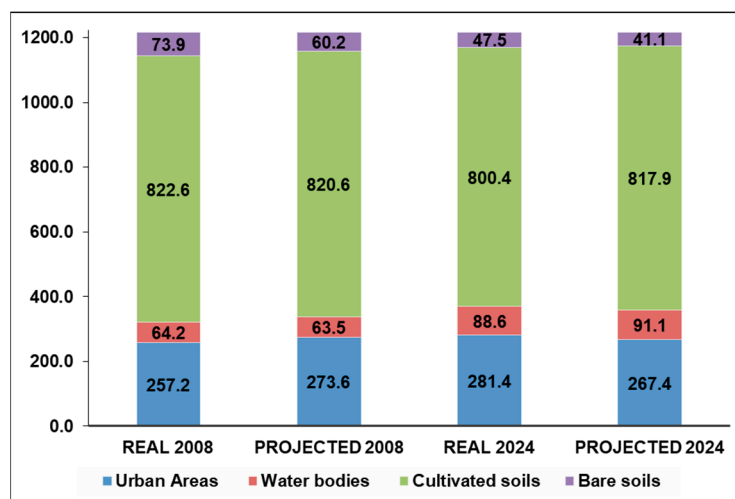


Figure 11. The comparison between simulated and real in km² of 2008 and 2024.

3.7. Limitations and Prospects

Since the training dataset was gathered for a particular geographic location, altering the study area may have an impact on the model's performance, even though a comprehensive training dataset with adequate diversity (covering all land types) and encompassing different pixel-level and object-level features for training classifiers was collected. However, if training samples are provided from the new target regions, the framework can be successfully applied in additional study areas. There are a number of difficulties when working with satellite imagery, such as the existence of biases and insufficient information. For

example, in the target study area, cloud coverage may impede the use of important data for LULC mapping. Furthermore, depending on just one distant sensing source may present difficulties due to complex classes within the LULC system. In order to overcome the limits posed by insufficient satellite imagery and intricate class distinctions, it is advised that additional data sources be incorporated and integrated in order to greatly increase the quality and dependability of LULC mapping. This study ran into a number of issues when utilizing GEE for classification despite its strong points, which include a broad range of features for spatial data processing, analysis, and interpretation. These restrictions include the platform's integration of different features, data processing volume, and computing time constraints. To guarantee the completeness of the training data, the model training procedure necessitates a large number of sample points when the study region is large. Using deep learning techniques and comparing their results is advised to generate dynamic LULC maps.

4. Conclusions

Using the classification and regression tree (CART) method on the Google Earth Engine (GEE) platform in the Qena Governorate, Egypt, this study concentrated on extracting LULC data from 1993 to 2024. Additionally, the CA-Markov model was utilized to predict future changes in LULC for the research area in 2040 and 2050. Urban areas grew by 57 km² between 1993 and 2008, while cultivated and bare soils shrank by 42.7 and 12.4 km², respectively. There was also a net loss of 1.9 km² in the Nile River. Water bodies expanded by 24.4 km², urban areas expanded by 24.2 km², and bare and farmed soils decreased by 26.2 and 26.4 km², respectively, between 2008 and 2024. Thematic maps of CA-Markov model showed the spatial distribution of anticipated expected increases in LULC for 2040 and 2050. According to the findings, there will be an increase in aquatic bodies, agricultural land, and urban regions. But, as expected, during the research years, the extent of bare lands decreased.

Planning decisions increasingly demand an awareness of the spatiotemporal dynamics of land use/land cover changes (LULCCs) as well as their current quantification and forecasting. Conducting a thorough analysis on LULC using the GEE platform and the CA-Markov chain model is an effective way for tracking and assessing LULC in the Qena Governorate over three distinct time periods, i.e., 1993, 2008, and 2024. Furthermore, the model can be beneficial for forecasting the LULC for 2040 and 2050. Land-cover dynamics have a substantial impact on land management, urban planning, risk assessment, ecosystem services assessment, and disaster response, emphasizing the importance of monitoring these dynamics. Across all model stages, an accuracy of over 90% was attained. Thus, in this investigation, CA-Markov modeling yielded results that are both dependable and accurate, which is promising. Additionally, this study demonstrated that using GEE and the CART model to analyze remote sensing data produced insightful information about patterns and changes in land cover.

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